MIL-I-43385B
9 September 1988
SUPERSEDING
MIL-I-43385A
13 June 1975

ICE CREAM CABINETS, MECHANICALLY REFRIGERATED, DISPENSING, MOBILE, SELF ELEVATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers two sizes of mechanically refrigerated mobile, dispensing ice cream cabinets for prepackaged ice cream slices or ice cream cups.

1.2 Classification. The ice cream cabinets covered by this document shall be of the following sizes as specified (see 6.2).

Size 1	- 450 standard 3 ounce ice cream cups or 4 ounce ice cream slices (minimum capacity)
Size 2	- 900 standard 3 ounce ice cream cups or 4 ounce ice cream slices (minimum capacity)

* 1.2.1 Military part number. The military part number shall consist of the letter M, the basic number of the specification, the size number, and the designations for lids, baskets, and bumpers as follows:

M43385 - 1 A 1 C
- M43385 — Document identification number.
- 1 — Size designation: 1 or 2
- A — Lid designation: A- opaque; B- clear
- 1 — Basket designation: 1- with; 2- without
- C — Bumper designation: C- with; D- without

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4110

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-391	-	Plastic Sheets, Rods and Tubing, Rigid Cast, Methacrylate (Multiapplication)
BB-F-1421	-	Fluorocarbon Refrigerants
* QQ-S-766	-	Steel, Plates, Sheets, and Strip - Corrosion Resisting
LLL-B-810	-	Building Board, (Hardboard) Hard Pressed, Vegetable Fiber

MILITARY

MIL-R-3065	-	Rubber, Fabricated Products
MIL-R-12323	-	Refrigerators and Related Equipment, Packaging and Packing of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	-	Identification Marking of US Military Property

* (Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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UNDERWRITERS LABORATORIES (UL)

Standard No. 471 - Commercial Refrigerators and Freezers

* (Copies may be obtained from the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

* NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - The National Electrical Code

(Copies may be obtained from the National Fire Protection Association, Batterymarch Park, Quincy, MA 02169.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 1 - Soda Fountain, Luncheonette Equipment

Listing of Food Service Equipment

* (Copies may be obtained from the National Sanitation Foundation, 3475 Plymouth Road, PO Box 1468, Ann Arbor, MI 48106.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR CONDITIONING ENGINEERS (ASHRAE)

Standard No. 13 - Home Freezers, Methods of Rating and Testing

(Copies may be obtained from the American Society of Heating, Refrigerating and Air Conditioning Engineers, Inc., United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 269 - Seamless and Welded Austenitic Stainless Steel Tubing for General Service

B 117 - Salt Spray (Fog) Testing

D 395 - Rubber Property - Compression Set

* D 638 - Tensile Properties of Plastic

D 702 - Cast Methacrylate Plastic Sheets, Rods, Tubes, and Shapes

* D 2000 - Rubber Products in Automotive Applications, Standard Classification Systems for

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D 2287 - Nonrigid Vinyl Chloride Polymer and Copolymer Molding and Extrusion Compounds

* D 3951 - Standard Practice for Commercial Packaging

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

* (Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Standard product. The ice cream cabinets delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

* 3.3 Codes and standards. The ice cream cabinets shall comply with the applicable requirements of Underwriters Laboratories (UL) Standard No. 471, National Sanitation Foundation (NSF) Standard No. 1 (see 4.5) and with the National Fire Protection Association (NFPA) Standard No. 70 (see 3.5.3).

3.4 Materials. The materials used in the ice cream cabinets shall conform to the applicable requirements specified herein. Materials not definitely specified shall be of the quality normally used in the ice cream cabinets by the manufacturer. Materials that would cause toxic reaction or transfer flavor or odor to food being refrigerated shall not be used (see 6.4).

* 3.4.1 Stainless steel. Stainless sheet steel shall conform to the chemical composition of any AISI steel number in the 300 series specified in QQ-S-766. Stainless steel tube shall conform to the requirements of any type of the 300 series specified in ASTM A 269.

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3.4.2 Refrigerant. The refrigerant shall conform to type 12, 22 or 502 of BB-F-1421.

* 3.4.3 Composition board. Composition board shall conform to type II surface 1, finish d, design a of LLL-B-810.

3.4.4 Plastic sheet. Plastic sheet shall be clear and transparent and shall conform to the requirements of either type I, grade c of L-P-391 or type II, grade 3 of ASTM D 702.

* 3.5 Design and construction. The ice cream cabinets shall consist of an insulated enclosure with lids located on top to give access to a refrigerated storage compartment and fixed tray rails. The storage compartment shall have a self-elevating mechanism which maintains ice cream containers at a constant top level. The ice cream cabinets shall have four casters to provide mobility. An automatic self-contained refrigeration system shall be provided and shall maintain the ice cream cabinets at any temperature selected between minus 10°F and minus 15°F. The ice cream cabinets shall be of two sizes, size 1 having a minimum storage capacity of 450 standard 3 ounce ice cream cups or 4 ounce ice cream slices and size 2 having a minimum storage capacity of 900 standard 3 ounce ice cream cups or 4 ounce ice cream slices.

3.5.1 Cabinet. The cabinet shall form a completely rigid structure to provide support to a full load of ice cream and the refrigerating system. Metal framing members shall be galvanized or phosphatized, and coated with a zinc chromate primer. There shall be no continuous metal connection between the outer shell and inner liner except as required by UL Standard No. 471. The overall maximum dimensions of the ice cream cabinets shall be as shown in table I.

TABLE I. Overall dimensions 1/

Size	Width (in. max.)	Length (in. max.)	Height (in. max.)
1	32-1/2	56	37
2	32-1/2	72	37

1/ Overall dimensions include accessories such as casters, bumpers and handles.

* 3.5.1.1 Outer shell. The outer shell shall be fabricated from 18 gage (.048) inch stainless steel as specified in 3.4.1. Joints shall be spot or seam-welded where practicable. Joints that are not welded air tight shall be sealed to prevent moisture infiltration into the insulation.

* 3.5.1.2 Inner liner. The sides and bottom of the inner liner shall be fabricated from 22 gage (.029) inch stainless steel as specified in 3.4.1. Seams shall be locked and soldered, or locked, spot welded, and mastic sealed to provide a leak proof seam.

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* 3.5.1.3 Top. The top shall be fabricated from one piece of 18 gage (.048) inch stainless steel as specified in 3.4.1. The top shall be insulated and sealed and shall be reinforced to prevent permanent deformation when foamed in place insulation is not used.

* 3.5.1.4 Breaker strips. The breaker strips shall be fabricated of composition board as specified in 3.4.3, or styrene plastic, or vinyl-chloride plastic, or a combination of these materials. Where foamed-in-place polyurethane insulation is used, all plastics in direct contact with the insulation shall be ABS (acrylonitrile, butadiene, styrene). The breaker strips shall join and shall make a vapor seal with the inner liner and outer shell. The edges of the breaker strips shall be eased or rounded. Styrene-type plastics shall have a minimum Izod impact resistance of 0.7 foot-pounds when tested as specified in 4.6.1.4. All plastics used for breaker strips shall not impart odor or taste to foods when tested as specified in 4.6.1.5.

* 3.5.1.5 Lids. Unless otherwise specified (see 6.2), the lids shall be fabricated of the stainless steel specified in 3.4.1 or of molded fiberglass. The lids shall be insulated and shall have gaskets as specified in 3.5.1.10 or other provisions to form a tight seal between the lids and the cabinets. The lids shall be equipped with lifting handles. When specified (see 6.2), the lids shall be the see-through type fabricated of the plastics specified in 3.4.4. The lid shall provide a tight seal between the lid and the cabinet.

3.5.1.6 Self-elevating mechanism. Each integral storage compartment shall be equipped with a drop in type self-elevating mechanism or dispenser which can be manually adjusted and maintain ice cream at a constant top level when tested as specified in 4.6.7. All materials used in the self elevating mechanism shall be corrosion-resistant.

3.5.1.7 Baskets. Unless otherwise specified (see 6.2), baskets having a height of 4 inches and fabricated from aluminum, stainless steel or corrosion-resistant metal, shall be furnished with each ice cream cabinet. The number of baskets shall be equal to the number necessary for full capacity operation of the cabinet.

3.5.1.8 Casters. Four casters shall be furnished on each cabinet and shall be bolted to gusset plates at the bottom of the base. The casters shall be full swivel type and shall have hard rubber or canvas-phenolic resin impregnated wheels having a minimum diameter of 5 inches. A minimum of two casters shall be equipped with a foot operated braking device which shall hold the cabinet stationary when tested as specified in 4.6.5. Each of the four casters shall be capable of supporting a load of 250 pounds without failure. The casters shall not be thrown out of alignment nor mark the floor when tested as specified in 4.6.5.

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3.5.1.9 Insulation. The insulation shall be industrial-type fiberglass insulation board, expanded polystyrene, or preformed or foamed-in-place fluorocarbon expanded polyurethane. The insulation shall be homogeneous throughout and shall have a "K" factor not greater than the specified values at a 75°F mean temperature. The "K" factor rating shall not be more than 0.25 British thermal units (Btu) per hour, per square foot, per degree F temperature differential, per inch of thickness for fiberglass insulation board and expanded polystyrene and not more than 0.15 for expanded polyurethane. Loose fill type insulation shall not be acceptable. The lids and the spaces between the outer shell and inner liner shall be completely filled with any type insulation specified above. Provisions shall be made to prevent moisture from entering the insulation of the cabinets by having the evaporator or part of the evaporator bonded to the insulation side of the inner liner, or having tubing or other refrigerant containing parts, subject to temperatures below 32°F, in contact with the insulation.

3.5.1.9.1 Navy. For Navy procurement (see 6.2), the manufacturer's standard commercial insulation shall be used.

* 3.5.1.10 Gaskets. Gaskets shall be rubber, synthetic rubber, or vinyl-chloride plastics. The material for rubber gaskets shall conform to M3BC 507 A14E034 of ASTM D 2000 and MIL-R-3065. Vinyl-chloride plastic shall have a compression set of less than 20 percent when tested as specified in 4.6.1.1, a minimum tensile strength of 1100 p.s.i. with minimum elongation at rupture of 250 percent when tested as specified in 4.6.1.2, and a weight loss not exceeding 2-1/2 percent when tested as specified in 4.6.1.3. The gaskets shall be replaceable cushion-type, one-piece units and shall be symmetrically installed to provide air-tight sealing.

3.5.1.11 Outer shell temperature. The temperature of the outer shell of the cabinets shall not be more than 10°F below the ambient temperature when the refrigeration system is in operation and the temperature in the storage compartment is being held at 0°F when tested as specified in 4.6.2.

3.5.1.12 Handles. The cabinet shall be provided with push-pull handles at both ends.

* 3.5.1.13 Bumpers. Unless otherwise specified (see 6.2), wrap around bumpers shall be provided with each cabinet. The bumpers shall extend around the full perimeter of the cabinet base. Material for bumpers shall be as specified in 3.5.1.10 or that which is normally supplied by the manufacturer.

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* 3.5.1.14 Tray rails. When specified (see 6.2), a fixed type tray rail consisting of three 1 inch nominal diameter stainless steel tubes specified in 3.4.1 with supporting hardware shall be supplied with each ice cream cabinet. Tray rails shall be mounted on either the compressor vent side or on the other side as specified (see 6.2). The tray rail shall extend the full length of the cabinet and the ends of each tube shall be closed. The height of the tray rail from the floor shall be adjustable from 34 to 35 inches. The tray rails shall be capable of supporting the weights listed below equally distributed over their full length with no evidence of deformation when tested as specified in 4.6.4.3:

Size 1 - 75 pounds
Size 2 - 85 pounds

3.5.2 Refrigeration system. The refrigeration system and component parts shall be completely installed and connected in the machine compartment to make a fully automatic, self-contained, mechanically cooled ice cream cabinet ready for connection to an external electric power source.

3.5.2.1 Condensing unit. The condensing units shall be a hermetic type, thermally or mechanically sealed. The hermetic condensing units shall use the refrigerant specified in 3.4.2.

3.5.2.2 Evaporator. The evaporator shall be copper tubing fastened to the insulation side of the inner liner, or shall be refrigerated dividers in the cabinet, or a combination of both. A thermal mastic, an asphalt material, or solder shall be used to fillet the tubing on the liner for maximum efficiency. The tubing shall not be placed where it will cause frost accumulation on the breaker strips.

* 3.5.2.3 Temperature control. The ice cream cabinets shall be equipped with a thermostat to maintain temperature control. The control shall be adjustable to maintain temperatures selected between minus 10°F and minus 15°F. The thermostat shall be adjusted and installed so that the cabinet temperature will not vary more than plus or minus 5°F from the temperature setting when ambient conditions vary from 50°F to 110°F when tested as specified in 4.6.6.2. An external dial type thermometer, calibrated in F and having a temperature scale range which includes the operating range of the cabinet, shall be furnished.

* 3.5.2.4 Refrigerant charge. The assembled refrigeration system shall be evacuated, completely dehydrated, and charged with the necessary amount and type of refrigerant and lubricating oil necessary for operation.

* 3.5.2.5 Performance. The refrigeration system, when installed as part of a complete ice cream cabinet, shall be capable of undergoing a no-load pull-down, a no-load, and a storage load test as specified in 4.6.6. A temperature of minus 5°F shall be attained in not more than 8 hours in an ambient of 110°F for the no-load pulldown test. The running time for the no-load test shall not exceed 60 percent over a 6-hour period. The running time for the storage load test shall not exceed 65 percent over a 6-hour period. The ambient conditions

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for the no-load and storage load test shall be 90°F and the stabilized freezer temperature at the beginning of the tests and during the tests shall be minus 5°F plus or minus 2°F. The ice cream cabinets when specified for 50 Hertz (Hz) operation (see 6.2), shall be tested as above when operating on a 50 Hz current at the motor manufacturer's rated voltage except that the running time for the storage load test shall not exceed 75 percent. The ice cream cabinets shall be tested with the current characteristics specified for the unit in the acquisition document. The ice cream cabinets shall not cut out on overload during these tests.

* 3.5.3 Electrical requirements. Unless otherwise specified (see 6.2), ice cream cabinets shall be designed for operation on a 115-volt, 60 Hertz (Hz) single phase system. All ice cream cabinets shall be completely electrically wired and shall conform to NFPA 70. Motors shall be protected against damages from overloads and voltage fluctuations when starting or running. When permitted by UL 471, a power supply cord shall be furnished as part of the ice cream cabinet. The cord shall have an attachment plug and shall extend at least 6 feet, but not more than 10 feet beyond the point at which it is secured to the back of the cabinet. The power supply cord shall be three wire with a three prong grounding plug.

3.5.4 Structural rigidity. The complete ice cream cabinets, with casters, shall show no damage or permanent deformation of any part of the cabinet, base, or refrigeration system when tested as specified in 4.6.4.1 and 4.6.4.2.

* 3.6 Finish. All stainless steel surfaces shall be ground and polished to a No. 4 finish and finished in accordance with the manufacturer's standard practice.

3.6.1 Corrosion. There shall be no visible corrosion of the outer shell or inner liner when tested as specified in 4.6.3.

3.7 Operation. Each ice cream cabinet shall be tested as specified in 4.6.8 to determine that the cabinets operate properly and that the refrigeration system cycles properly. There shall be no refrigerant leaks and the moisture content of the refrigerant at 100°F shall not exceed 10 parts per million for the refrigerant specified in 3.4.2 when tested as specified in 4.6.8.

3.8 Inspection after testing. After testing, the cabinets shall be allowed to come to equilibrium with room temperature. The interior shall be dried of any condensation and thoroughly aired and inspected.

* 3.9 Data name plate. The ice cream cabinets shall be furnished with a data name plate in accordance with the applicable requirements of MIL-STD-130 except the requirements for; (a) Methods of applying, (b) Identification tags, (c) Information not required, and (d) Optional marking information shall not apply.

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The data name plate shall be made of minimum 20 gauge (.036) inch corrosion resisting metal, and attached to cabinets by rivets or screws in such a manner as to meet the applicable requirements for this type of equipment. The plate shall bear the following information which shall be stamped, engraved, or applied by photosensitive means:

National Stock Number
Procurement Contract Number
Specification Data
Manufacturer's Name, Address, and Telephone Number
Supplier's Name, Address, and Telephone Number (If different from manufacturer.)
Manufacturer's Model Number
Government Approved Manual Number, when specified (see 6.2)

Each plate shall be placed so as to be readily visible to the operator during normal operation and use, and so as to not adversely affect the life or utility of the cabinets.

* 3.10 Workmanship. All components and assemblies of the ice cream cabinets shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, flashing, pits, blow holes and sprues. External surfaces shall be free from sharp edges and corners, except when corners are required. The cabinets shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

3.10.1 Metal fabrication. Metal used in the fabrication of the ice cream cabinets shall provide for original quality surface finish and shall be free from kinks and unspecified bends. Forming and shearing shall not cause damage to the metal and shall be done neatly and accurately. Corners shall be square and true and all bends of a major nature shall be made with dies or fixtures to insure uniformity of size and shape.

3.10.2 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be smooth and free of cracks, burn holes, undercuts or incomplete fusion. All scale and flux shall be removed from the finished weld area.

3.10.3 Fastening devices. Threaded fastener and rivet holes shall be accurately punched or drilled and shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be securely torqued. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested for the characteristics specified in 4.6.1 through 4.6.8. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

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* 4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Component testing. The plastic tests shall be performed as specified in 4.6.1.1 through 4.6.1.5. The inspection lot shall be the amount of each type of plastic offered for inspection at one time. Any nonconformance shall be cause for rejection of the lot.

* 4.4.2 In-process inspection. Inspection shall be made of the following operations or assemblies to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to the items affected and to the operation. Items which cannot be corrected shall be removed from production.

a. Galvanizing or phosphatizing, and priming of metal framing members is in compliance with 3.5.1.

b. Sealing of opening in outer shell is in compliance with 3.5.1.1.

c. Sealing of inner liner seams is in compliance with 3.5.1.2.

d. Reinforcing of top, when applicable, is in compliance with 3.5.1.3.

e. Installation of insulation is in compliance with 3.5.1.9.

f. Filleting of tubing is in compliance with 3.5.2.2.

g. Charging of refrigeration system is in compliance with 3.5.2.4.

* 4.4.3 End item visual examination. The ice cream cabinets shall be examined for the defects listed in table II. The lot size shall be expressed in units of cabinets. The sample unit shall be one cabinet. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Not manufacturer's standard finish		X
Construction and workmanship			
Breaker strips	Edges not eased or rounded		X
Lid	No lifting handles	X	
	Not see through type (when specified)	X	
Tray rail (when specified)	Missing, mounted on wrong side, not tubular construction, not fixed type	X	
Self-elevating mechanism	Cannot be removed	X	
	Cannot be manually adjusted	X	
Baskets	Not provided (when applicable)	X	
Casters	Incorrect number	X	
	Not bolted to gusset plates		X
	Not swivel type	X	
	No braking device on minimum of two casters	X	
	Braking device not foot operated		X
Liquid indicator	Missing (when applicable)	X	
Thermostat	Not adjustable	X	
Thermometer	Not external dial type		X
Gaskets	Not one piece	X	
	Not replaceable	X	
	Not as specified	X	
Handles	Not located at both ends		X

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Bumpers (when applicable)	Do not extend around cabinet base perimeter		X
Data name plate	Omitted or not as specified		X
	Information incomplete or illegible		X
	Not located so as to be readily visible to the operator		X

* 4.4.4 End item dimensional examination. The end item shall be examined to determine compliance with the dimensions specified in section 3. Any nonconformance shall be considered a defect. The lot size shall be expressed in units of cabinets. The sample unit shall be one cabinet. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.5 End item testing. Each ice cream cabinet shall be tested as specified in 4.6.8. Any nonconformance shall be cause for rejection of the unit. One representative production unit (see 6.3) shall be tested as specified in 4.6.2 through 4.6.7. Any nonconformance shall be cause for rejection of the inspection lot.

* 4.4.6 Packaging inspection. The inspection shall be in accordance with the quality assurance provisions of MIL-R-12323 or ASTM D 3951, as appropriate.

* 4.5 Code and standard compliance. Certifications, certified test reports, or listing marks for codes and standards, as applicable, submitted in accordance with 3.3, shall be examined and validated as proof of compliance. Prior to Government approval of the first shipment, the contractor shall submit to the contracting officer or his authorized representative for approval, certification that the ice cream cabinets he proposes to furnish meet the applicable requirements of NSF and UL as follows:

* 4.5.1 Underwriter's Laboratories, Inc. (UL). Acceptable evidence of meeting the applicable requirements of the Underwriter's Laboratories standards specified in 3.3 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent laboratory, acceptable to the Government, indicating that the ice cream cabinet conforms to the applicable requirements of the specified UL standards.

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* 4.5.2 National Sanitation Foundation (NSF). Acceptable evidence of meeting the applicable requirements of the National Sanitation Foundation standards specified in 3.3 shall be one of the following.

a. A listing in the current edition of the National Sanitation Foundation "Listing of Food Service Equipment", and display of the NSF seal on the finished unit.

b. A certification for the unit issued by NSF under their special onetime contract evaluation/certification service.

c. A certified test report acceptable to the contracting officer with the advice of the Army Surgeon General, from an independent testing laboratory, indicating that the unit has been tested and conforms to the specified NSF standards.

* 4.6 Methods of inspection.

4.6.1 Plastic tests.

4.6.1.1 Compression set test. A one inch section of vinyl-chloride shall be compressed in the direction of normal use to 60 percent of the original free height in a device described in method B of ASTM D 395 and held for 7 days at 70°F plus or minus 2°F. The free height shall be measured 30 minutes after removal from the test block to determine if the compression set is within the limits specified in 3.5.1.10. Failure to meet the compression requirements of 3.5.1.10 shall constitute failure of this test.

* 4.6.1.2 Tensile strength test. Vinyl-chloride plastic shall be tested for tensile strength in accordance with ASTM D 638 to determine compliance with 3.5.1.10. Failure to meet the tensile strength requirement of 3.5.1.10 shall constitute failure of this test.

4.6.1.3 Weight loss test. Vinyl-chloride shall be tested for weight loss in accordance with test 8 (g) of ASTM D 2287 to determine compliance with 3.5.1.10. Failure to meet the weight loss requirements of 3.5.1.10 shall constitute failure of this test.

* 4.6.1.4 Impact resistance. Styrene-type plastic shall be tested for Izod impact resistance in accordance with ASTM D 638 to determine compliance with the impact resistance requirement of 3.5.1.4. Failure to meet the impact resistance requirement of 3.5.1.4 shall constitute failure of this test.

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4.6.1.5 Odor and taste test. Plastic shall be given the following odor and taste test to determine compliance with the odor and taste requirements of 3.5.1.4. Place the plastic to be tested with approximately 5 cc of distilled water in a 16 ounce, wide-mouth glass jar. Place a beaker containing 100 to 150 cc of distilled water in the jar so that the water does not touch the plastic being tested. The jar shall be closed tightly. Place 100 to 150 cc of distilled water in a second closed glass jar. The two jars shall be kept at 45°F to 50°F for 24 hours. At the end of this period, an examination shall be made for odor or taste difference between the two waters. Containers of other shapes, capacities and types may be substituted for the containers described if odd shaped pieces of plastic are tested, provided the intent of the odor and taste test is met. Failure to meet the odor and taste requirement of 3.5.1.4 shall constitute failure of this test.

4.6.2 Outer shell temperature test. The refrigeration system of a complete ice cream cabinet shall be operated in an ambient temperature of 90°F, plus or minus 2°F until the temperatures in the storage compartment stabilizes at 0°F, plus or minus 2°F. The temperature of the outer shell shall then be measured at five locations (front, back, top and sides) to determine compliance with 3.5.1.11. Failure to meet the outer shell temperature requirements of 3.5.1.11 shall constitute failure of this test.

4.6.3 Salt spray test. The interior and exterior surfaces of the ice cream cabinets shall be subjected to a salt spray for 48 hours in accordance with ASTM B 117 to determine compliance with 3.6.1. Sample 6 inch by 12 inch panels, typical of the inner liner and outer shell materials, finish and trim, may be subjected to the salt spray test in lieu of the complete cabinet. Failure to meet the salt spray requirements of 3.6.1 shall constitute failure of this test.

4.6.4 Structural rigidity tests.

4.6.4.1 Rigidity test. The complete ice cream cabinet shall be tilted on one side bottom edge so that the bottom makes an angle of 45 degrees with the horizontal to determine compliance with 3.5.4. The test shall be repeated with the opposite bottom edge bearing the load. Failure to meet the sturdiness requirement of 3.5.4 shall constitute failure of this test.

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4.6.4.2 Drop test. The complete ice cream cabinet shall be loaded to 50 percent of normal capacity by volume with suitable materials not in liquid form, having an overall density of 35 pounds per cubic foot to determine compliance with 3.5.4. The material used for loading shall be nonrigid or shall consist of increments. The load shall be distributed uniformly over the floor of the liner. One side of the ice cream cabinet at a time shall then be tilted 3 inches off a level concrete floor and shall be allowed to fall freely in the following order: Left side, rear, right side, front. The ice cream cabinet shall be held in the tilted position for the drop test by a single support which can be withdrawn so the ice cream cabinet is not hampered in its free fall or caused to fall in a plane other than parallel to the specified direction of fall. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

* 4.6.4.3 Tray rail weight bearing test. The tray rail on each size cabinet shall be tested for compliance to the weight bearing requirements of 3.5.1.14. The required weight for the size being tested shall be equally distributed over the full length of the tray for a full ten minutes before being removed. Any evidence of bending, distortion, deflection or other form of deformation of the tray rail after removal of the weight, shall constitute a test failure.

4.6.5 Caster test. An evenly distributed load, not in liquid form, shall be placed in the storage compartment of the ice cream cabinet. The load placed in the storage compartment plus the weight of the cabinet shall equal 1000 pounds. The cabinet, with casters, shall be rolled back and forth over a smooth, clean, concrete floor and a 3/4 inch beveled board for 2 minutes. During the test the casters shall be observed to be able to swivel and roll smoothly. Following the test, the casters and floor shall be examined to determine compliance with the failure, alignment and marking requirements of 3.5.1.8. The cabinet shall then be placed at the top slope of a ramp, having a minimum incline of 20 degrees, with the brakes locked for one minute to determine compliance with the braking requirement of 3.5.1.8. Failure to meet caster requirements of 3.5.1.8 shall constitute failure of this test.

4.6.6 Performance preparation and test.

* 4.6.6.1 Performance preparation.

a. The refrigeration system for the ice cream cabinet shall be evacuated, charged, filled with refrigerant oil and operated for a period of approximately 8 hours at room temperature. Controls shall be checked and the power supply voltage shall be plus or minus 5 volts of the voltage specified for operation. Other electrical requirements shall be as specified.

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b. Air circulation in the test room shall maintain the specified uniformity of temperature, but the cabinet shall be shielded from forced air currents. Ambient temperatures shall be measured at points 1 foot above the floor and 3 feet above the cabinet at the center point on three sides. Ambient temperatures at each point shall be maintained within plus or minus 2°F of the specified test conditions. The test conditions shall be maintained during stabilization periods as well as during test runs.

c. Ambient temperatures shall be measured with thermocouples, resistance thermometers, or glass thermometers. The temperature sensitive part of the temperature measuring device shall be in good thermal contact with a metallic mass having an equivalent total heat capacity of not less than 10 or more than 20 grams of water.

d. In preparation for any operating test, the ice cream cabinet shall be operated for the period specified to obtain steady state conditions. Immediately following the preparatory test period, the cabinet shall be operated for the performance test period during which the elements of performance are determined and recorded. Temperature observations shall be made continuously or at equal intervals of not greater than 15 minutes. Reported temperatures shall be the arithmetical average of all readings taken at equal time intervals.

e. The ice cream cabinet shall be loaded with a simulated load of ice cream corresponding to 100 percent of the specified load.

f. The interior ice cream cabinet temperature shall be measured with not less than eight temperature measuring devices, equally distributed throughout the load for the storage load test, and equally distributed throughout the cabinet for no-load tests.

g. The condensing unit of the ice cream cabinet shall be placed in operation when the ambient temperature, determined as the average of the temperature measuring devices on the outside of the cabinet, is as specified. During the test period, observations shall be made that the interior ice cream cabinet temperatures determined as the average of the temperature measuring devices located in the cabinet, shall be as specified. No single interior temperature shall exceed 0°F.

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4.6.6.2 Performance test. A no-load pull-down test, a no-load test, and a storage load test shall be performed on the ice cream cabinets according to ASHRAE Standard No. 13 and the requirements of this document to determine compliance with 3.5.2.5. The thermostat control shall be checked for compliance with 3.5.2.3 by operating the ice cream cabinet with a temperature control setting of 0°F in a steady state ambient condition of 50°F and after the cabinet temperature has stabilized and held at the control setting temperature over a 2 hour period. The ambient temperature shall be raised at an approximately constant rate to 110°F over a 6 hour period. The storage load test for refrigeration units procured for operation on a frequency of 50 cycles shall be performed by operating for 4 hours at the specified frequency and at 90 percent of the specified system voltage followed by an operating test of 4 hours at specified frequency and at 110 percent of specified system voltage. Failure to meet the performance requirements of 3.5.2.3 and 3.5.2.5 shall constitute failure of this test.

4.6.7 Self-elevating mechanism test. The storage compartment of the ice cream cabinet shall be loaded to capacity with standard 3 ounce cups or 4 ounce slices of ice cream in baskets. The ice cream cups and baskets shall be removed in a serving line manner and the self-elevating mechanism shall be observed to determine compliance with 3.5.1.6. Failure to meet the self-elevating requirements of 3.5.1.6 shall constitute failure of this test.

* 4.6.8 Operation test. Each ice cream cabinet shall be operated long enough at normal control setting to determine that the cabinet and refrigeration system operates and cycles properly as specified in 3.7. Observation shall be made to insure motors are protected against fluctuation or damage (see 3.5.3). After completion of this portion of the test, and with the cabinets at 70°F, plus or minus 5°F, all refrigerant joints shall be checked for leaks in accordance with the manufacturer's standard method to determine compliance with 3.7. The test for leakage shall be made with an electronic leak detector that is capable of measuring a leakage rate of at least 1/2 ounce per year as minimum. Where the combination liquid and moisture indicator is used, it shall indicate "dry" within the limits specified in 3.7 during and upon completion of this test. Where there is no moisture indicator the moisture content of the refrigerant shall be determined in accordance with the manufacturer's standard method to determine compliance with 3.7. Failure to meet the requirements of 3.7 shall constitute failure of this test.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

* 5.1.1 Level A. Each ice cream cabinet shall be preserved in accordance with the applicable requirements of MIL-R-12323.

* 5.1.2 Commercial. Each ice cream cabinet shall be preserved in accordance with ASTM D 3951.

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* 5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

* 5.2.1 Levels A and B packing. Each ice cream cabinet, preserved as specified in 5.1, shall be packed in accordance with the applicable requirements of MIL-R-12323. The base of each shipping container shall be provided with cross loadbearing members to provide a minimum 1/2 inch clearance for casters.

* 5.2.2 Commercial packing. Ice cream cabinets, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-R-12323 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The ice cream cabinets are intended for use as mobile ice cream dispensers for mess halls.

* 6.2 Ordering data. Acquisition documents should specify the following.

a. Title, number and date of this document.
b. Size of ice cream cabinet required (see 1.2).
c. When a first article is required (see 3.1, 4.3 and 6.3).
d. When see-through type lids are required (see 3.5.1.5).
e. When lids are other than specified (see 3.5.1.5).
f. When baskets are not required (see 3.5.1.7).
g. When procurement is for Navy (see 3.5.1.9.1).
h. When wrap around bumpers are not required (see 3.5.1.13).
i. When tray rails are required and whether they shall be mounted on the compressor vent side or on the other side (see 3.5.1.14).
j. When 50 Hertz (Hz) electric motor is required, the contracting officer shall specify the motor voltage and phase (see 3.5.2.5 and 3.5.3).
k. When government approved manual number is to be supplied with data name plate (see 3.9).
l. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

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* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.4).

* 6.5 Subject term (key word) listing.

Cabinets
Food service equipment
Ice cream
Refrigeration equipment

* 6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the previous issue.

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Air Force - 82

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DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-43385B	ICE CREAM CABINETS, MECHANICALLY REFRIGERATED, DISPENSING, MOBILE, SELF ELEVATING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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